

Model-Aware Positive Stieltjes Quadrature: From Structural Minimax to Gauss–Legendre

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Abstract

Generalized Gaussian and rationally exact quadrature already exploit non-polynomial structure, while constrained rational minimax approximation is an established approximation problem. We study positive symmetric Stieltjes-response rules that minimize relative error uniformly over a continuum of Lorentzian widths while matching a variable prefix of polynomial moments. Within this class, n positive node pairs and moment depth k give the feasible range $0 \leq k \leq 2n$, admissible dimension $2n - k$, and the Gauss–Legendre endpoint at $k = 2n$. For $k < 2n$, a response-difference degree bound yields a $2n - k + 1$ -point sufficient alternation certificate. High-precision hierarchies for $n = 2, 3, 4$ quantify the trade-off between structural accuracy and polynomial protection. For six upper-half levels, Arb/Krawczyk calculations prove a unique exact zero of each fixed-sign moment–equioscillation system inside an explicit radius- 10^{-12} box; interval derivative and curvature tests further prove that the alternating level is the global uniform relative-error norm on the structural interval. For a smooth residual with Chebyshev coefficients $|a_j| \leq C\rho^{-j}$, an exact coefficient-space norm gives the model-aware risk $G_k(\rho) + \tau S_k$. Ball arithmetic encloses all nine structural maxima and the Chebyshev defects of the rounded $n = 4$ operators. Adaptive interval comparison certifies a unique selector on 99.5956% of the tested $(\rho, \log_{10} \tau)$ -coordinate area, leaving 0.4044% unresolved near transition curves. A separate forward-automatic-differentiation/Arb code path reproduces all six local root certificates, while repository-wide regression checks reproduce the full 21-rule hierarchy and the interval selector calculation.

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1. Introduction

Polynomial exactness is a central organizing principle of numerical quadrature, but it is not the only way to spend a fixed node budget. When the dominant difficulty is a nearby pole, quadrature may instead incorporate rational functions chosen to match that structure [1, 2]. Generalized Gaussian quadrature extends positive exactness to Chebyshev systems and is naturally described through moment-space geometry [3]. Positive interpolatory formulas and optimization-based moment-matching constructions provide additional general frameworks [7, 8, 9].

A second neighboring direction begins with rational approximation. Modern algorithms compute rational minimax approximants through adaptive barycentric representations and Remez-type iterations [4]. Interpolation-constrained rational minimax approximation, including higher-order constraints, is also established [5]. For Markov and Stieltjes functions, relative rational approximation has a substantial theory [10]. Most directly, quadrature formulas can be derived from rational approximations of Cauchy transforms, with poles and residues becoming nodes and weights and Gauss quadrature connected to Padé approximation at infinity [6].

These results rule out several broad novelty claims: pole-aware quadrature, positive exact quadrature, rational minimax, equioscillation, and constrained rational approximation are not new separately. The narrower problem studied here is to combine them in a model-aware design. At fixed node-pair count, we require a positive symmetric Stieltjes response, match a variable prefix of polynomial moments, and minimize relative error uniformly over a continuum of Lorentzian widths. The resulting family spans a structural minimax rule and the Gauss–Legendre endpoint. The practical question is then not only how to compute the rules, but how to select the moment depth when a smooth uncertain residual is superposed on the structural component.

Within this stated class, the paper makes the following contributions:

- (i) it formulates the positive variable-moment continuum-minimax problem and characterizes its feasible range, dimension, and Gauss–Legendre endpoint;

- (ii) it gives a short class-specific degree/zero-count argument leading to a $2n - k + 1$ -point alternation certificate for every nonzero-dimensional level;
- (iii) it computes the complete path for $n = 2, 3, 4$, including the upper-half levels between the earlier lower-half designs and the Gaussian endpoint;
- (iv) it derives an exact mixed structural/smooth risk and uses it to select the moment depth across a two-parameter model domain;
- (v) it gives computer-assisted existence and local uniqueness boxes for the six upper-half square systems and certifies their global alternation levels on the structural interval;
- (vi) it supplies a conservative interval-certified decision map, a shifted-pole scope test, and multi-start reconstruction from perturbed initializations;
- (vii) it reproduces the full 21-rule hierarchy and all six upper-half Krawczyk verdicts through a separate author-produced verification path using forward automatic differentiation over Arb intervals.

The feasible-range theorem and the Gaussian endpoint are important structural components of the framework, but are not claimed as stand-alone inventions of moment or generalized Gaussian theory. A targeted literature search did not identify a direct source combining positivity, continuum Lorentzian minimax, variable moment depth, an exact mixed selector, and an interval-certified parameter partition. This is not an exhaustive proof of novelty; specialist prior-art review remains necessary. The claims are restricted to the symmetric positive class and do not imply universal superiority over Gaussian, adaptive, or problem-specific quadrature.

2. Problem formulation

Let

$$I[f] = \int_{-1}^1 f(x) dx, \quad L_b(x) = \frac{1}{x^2 + b^2}, \quad b \in B = [0.08, 0.12].$$

Set

$$S = \{b^2 : b \in B\} = [0.08^2, 0.12^2].$$

With $s = b^2$ and $y = x^2$,

$$F(s) = I[L_{\sqrt{s}}] = \int_0^1 \frac{y^{-1/2}}{s+y} dy = \frac{2}{\sqrt{s}} \arctan \frac{1}{\sqrt{s}}, \quad s \in S.$$

For a fixed integer $n \geq 1$, define the ordered positive parameter domain

$$\mathcal{P}_n = \{(y, w) \in \mathbb{R}^{2n} : 0 < y_1 < \cdots < y_n < 1, \quad w_j > 0\}.$$

Each $(y, w) \in \mathcal{P}_n$ determines the symmetric rule

$$Q_{y,w}[f] = \sum_{j=1}^n w_j \{f(-\sqrt{y_j}) + f(\sqrt{y_j})\}.$$

Its response on the structural family is

$$r_{y,w}(s) = Q_{y,w}[L_{\sqrt{s}}] = 2 \sum_{j=1}^n \frac{w_j}{s + y_j}.$$

For an integer $k \geq 0$, let

$$m^{(k)} = \left(\frac{1}{2\ell + 1} \right)_{\ell=0}^{k-1}$$

and define the moment map

$$M_k(y, w) = \left(\sum_{j=1}^n w_j y_j^\ell \right)_{\ell=0}^{k-1}.$$

For $k = 0$, both vectors are empty. The admissible parameter set and the corresponding rule class are

$$\Theta_{n,k} = \{(y, w) \in \mathcal{P}_n : M_k(y, w) = m^{(k)}\}, \quad \mathcal{A}_{n,k} = \{Q_{y,w} : (y, w) \in \Theta_{n,k}\}.$$

Lemma 1 (Equivalent interpretations of the moment constraints). *Let $k \geq 1$ and $(y, w) \in \mathcal{P}_n$. The following statements are equivalent:*

- (a) $(y, w) \in \Theta_{n,k}$;
- (b) $Q_{y,w}$ is exact for every polynomial of degree at most $2k - 1$;

(c)

$$r_{y,w}(s) - F(s) = O(s^{-k-1}) \quad \text{as } s \rightarrow +\infty.$$

Proof. Symmetry gives exactness for odd polynomials. For even monomials,

$$Q_{y,w}[x^{2\ell}] = 2 \sum_{j=1}^n w_j y_j^\ell, \quad I[x^{2\ell}] = \frac{2}{2\ell+1}.$$

This proves the equivalence of (a) and (b).

For every fixed $m \geq 1$, uniformly for $y \in [0, 1]$,

$$\frac{1}{s+y} = \sum_{\ell=0}^{m-1} (-1)^\ell y^\ell s^{-\ell-1} + O(s^{-m-1}).$$

Consequently,

$$F(s) = 2 \sum_{\ell=0}^{m-1} \frac{(-1)^\ell}{2\ell+1} s^{-\ell-1} + O(s^{-m-1}),$$

whereas

$$r_{y,w}(s) = 2 \sum_{\ell=0}^{m-1} (-1)^\ell \left(\sum_{j=1}^n w_j y_j^\ell \right) s^{-\ell-1} + O(s^{-m-1}).$$

Taking $m = k$ proves that (a) implies (c). Conversely, if (c) holds, uniqueness of the coefficients in these finite asymptotic expansions forces equality of the first k moments, proving (c) implies (a). $\square$

Lemma 2 (Uniqueness of the representation). *If $(y, w), (\hat{y}, \hat{w}) \in \mathcal{P}_n$ and*

$$r_{y,w}(s) = r_{\hat{y},\hat{w}}(s)$$

on a nonempty open real interval, then

$$(y, w) = (\hat{y}, \hat{w}).$$

Proof. The identity theorem makes the two rational functions identical. Their simple poles are $-y_j$ and $-\hat{y}_j$, and the corresponding residues are $2w_j$ and $2\hat{w}_j$. Equality of poles and residues, together with the imposed ordering, gives equality of the parameter vectors. $\square$

3. Feasible-range characterization

Theorem 3 (Feasible range, regular dimension, and Gaussian endpoint). *For $n \geq 1$, the following statements hold:*

- (i) $\Theta_{n,k} \neq \emptyset$ for every integer $0 \leq k \leq 2n$;
- (ii) for $0 \leq k \leq 2n$, $\Theta_{n,k}$ is an embedded smooth submanifold of the open set $\mathcal{P}_n$, of dimension

$$\dim \Theta_{n,k} = 2n - k;$$

- (iii) $\Theta_{n,k} = \emptyset$ for every integer $k \geq 2n + 1$;
- (iv) $\Theta_{n,2n}$ is a singleton, and its associated symmetric rule is the classical $2n$ -point Gauss–Legendre rule.

Proof. Let $d\mu(y) = y^{-1/2} dy$ on $[0, 1]$. By the standard Gaussian quadrature theorem for a positive measure with infinite support, there exists a unique n -node rule with positive weights that is exact for all polynomials of degree at most $2n - 1$. Write its nodes and weights as y_j and $2w_j$. Then

$$\sum_{j=1}^n w_j y_j^\ell = \frac{1}{2} \int_0^1 y^\ell d\mu(y) = \frac{1}{2\ell + 1}, \quad 0 \leq \ell \leq 2n - 1.$$

Thus $\Theta_{n,2n} \neq \emptyset$, and every lower moment prefix is also feasible. The case $k = 0$ is simply $\Theta_{n,0} = \mathcal{P}_n$.

For $1 \leq k \leq 2n$, suppose a linear combination of the rows of $DM_k(y, w)$ vanishes, and define

$$p(t) = \sum_{\ell=0}^{k-1} c_\ell t^\ell.$$

The columns corresponding to w_j give $p(y_j) = 0$. The columns corresponding to y_j give $w_j p'(y_j) = 0$, and positivity gives $p'(y_j) = 0$. Hence every y_j is a double root of p . Since $\deg p \leq k - 1 \leq 2n - 1$, this is possible only when $p \equiv 0$. Therefore DM_k has row rank k at every point of $\mathcal{P}_n$. The regular-level-set theorem gives the stated embedded-manifold dimension.

If $k \geq 2n + 1$, the constraints include exactness for the transformed polynomial

$$g(y) = \prod_{j=1}^n (y - y_j)^2,$$

which has degree $2n$, is nonnegative, and is not identically zero. The discrete rule gives

$$2 \sum_{j=1}^n w_j g(y_j) = 0,$$

whereas

$$\int_0^1 g(y) d\mu(y) > 0,$$

a contradiction.

It remains to identify the endpoint uniquely. Let

$$\pi_n(y) = \prod_{j=1}^n (y - y_j)$$

for any element of $\Theta_{n,2n}$. For every polynomial q of degree at most $n - 1$, exactness through degree $2n - 1$ gives

$$\int_0^1 \pi_n(y) q(y) d\mu(y) = 2 \sum_{j=1}^n w_j \pi_n(y_j) q(y_j) = 0.$$

Thus π_n is the unique monic degree- n orthogonal polynomial for μ , so the nodes are uniquely determined. Exactness for degrees $0, \dots, n - 1$ then determines the weights uniquely through a nonsingular Vandermonde system. Under $y = x^2$, the resulting symmetric $2n$ -node rule is exact for all polynomials of degree at most $4n - 1$, and therefore is the $2n$ -point Gauss–Legendre rule. $\square$

Remark 4. The sets $\Theta_{n,k}$ are nested as moment constraints are added. The selected minimax rules form a discrete hierarchy indexed by k ; no connectedness of the level sets, and no continuous path of minimizers in k , is asserted.

4. Minimax characterization

For $(y, w) \in \mathcal{P}_n$, write

$$q_y(s) = \prod_{j=1}^n (s + y_j), \quad r_{y,w}(s) = \frac{p_{y,w}(s)}{q_y(s)},$$

where $\deg p_{y,w} \leq n - 1$.

Proposition 5 (Difference-numerator degree). *Let $Q_1, Q_2 \in \mathcal{A}_{n,k}$ with $0 \leq k \leq 2n$. Then either their responses are identical or*

$$r_{Q_1}(s) - r_{Q_2}(s) = \frac{H(s)}{q_1(s)q_2(s)}, \quad \deg H \leq 2n - k - 1.$$

For $k = 2n$, the second alternative is impossible.

Proof. The unreduced numerator

$$H = p_1q_2 - p_2q_1$$

has degree at most $2n - 1$. For $k = 0$, both responses are $O(s^{-1})$, so their difference is $O(s^{-1})$. For $k \geq 1$, the moment lemma gives

$$r_{Q_1}(s) - r_{Q_2}(s) = O(s^{-k-1}).$$

Since q_1q_2 is monic of degree $2n$, a nonzero polynomial numerator must satisfy

$$\deg H - 2n \leq -k - 1,$$

which is the claimed bound. □

Define

$$\varepsilon_Q(s) = \frac{r_Q(s)}{F(s)} - 1, \quad \|\varepsilon_Q\|_{\infty, S} = \max_{s \in S} |\varepsilon_Q(s)|.$$

Theorem 6 (Class-specific alternation certificate). *Let $0 \leq k < 2n$ and let $Q^* \in \mathcal{A}_{n,k}$. Suppose there are*

$$m = 2n - k + 1$$

distinct ordered points

$$s_1 < \cdots < s_m \quad \text{in } S$$

and a number $E > 0$ such that

$$\varepsilon_{Q^*}(s_i) = (-1)^i \sigma E, \quad \sigma \in \{-1, 1\},$$

and

$$E = \|\varepsilon_{Q^*}\|_{\infty, S}.$$

Then no rule in $\mathcal{A}_{n,k}$ has strictly smaller uniform relative error on S .

Proof. Assume that $Q \in \mathcal{A}_{n,k}$ satisfies

$$\|\varepsilon_Q\|_{\infty,S} < E.$$

At every s_i , the difference

$$\varepsilon_{Q^*}(s_i) - \varepsilon_Q(s_i)$$

has the sign of $\varepsilon_{Q^*}(s_i)$. These signs alternate, so continuity gives at least $m-1 = 2n-k$ distinct zeros in the open subintervals (s_i, s_{i+1}) . Since $F(s) > 0$ on S , these are zeros of $r_{Q^*} - r_Q$. The difference-numerator proposition allows at most $2n-k-1$ distinct zeros unless the responses are identical. In the identical-response case, uniqueness of the ordered positive representation gives $Q = Q^*$, which is incompatible with strict improvement. This contradiction proves the result. $\square$

Remark 7. The theorem is a sufficient global-minimax certificate within $\mathcal{A}_{n,k}$. It does not prove necessity of alternation, uniqueness among equal-norm minimizers, or existence of an exact equioscillating parameter vector from floating-point residuals. The endpoint $k = 2n$ is zero-dimensional and is fixed by its moments.

5. Numerical realization of the model-aware hierarchy

The nonlinear construction imposes exact moment equations, alternating equal-magnitude relative errors, and stationarity at internal extrema. We denote the ideal mathematical rule by $Q_{n,k}^*$ and its published decimal operator by $\tilde{Q}_{n,k}$. Lower-half rules were obtained first, and the upper-half problems were continued toward the Gaussian endpoint. The table and figures use the published decimal operators. For the six upper-half levels, Section 6 upgrades the floating-point roots to computer-assisted exact roots and certifies that their alternating levels are the global relative-error norms on the structural interval. The lower-half constructions remain high-precision numerical evidence rather than interval root certificates. All reported operators have ordered nodes and positive weights.

For $n = 4$, the error grows from 1.49×10^{-10} at $k = 0$ to 4.30×10^{-1} at the Gauss endpoint. The upper-half values $k = 5, 6, 7$ continue the approximately constant finite-range price per additional moment before the error enters an order-one regime. This pattern is empirical and is not claimed as an asymptotic theorem.

Table 1: Complete $n = 4$ hierarchy. The endpoint $k = 8$ is zero-dimensional; therefore the alternation column is not applicable.

k	Dimension	Exactness	Alt. points	Worst relative error	Min. w_j
0	8	—	9	1.490623e-10	0.039147
1	7	1	8	2.460626e-09	0.044528
2	6	3	7	4.048641e-08	0.051670
3	5	5	6	6.653546e-07	0.061646
4	4	7	5	1.092819e-05	0.076595
5	3	9	4	1.794333e-04	0.101407
6	2	11	3	2.943219e-03	0.149011
7	1	13	2	4.597982e-02	0.126194
8	0	15	—	4.300499e-01	0.101229

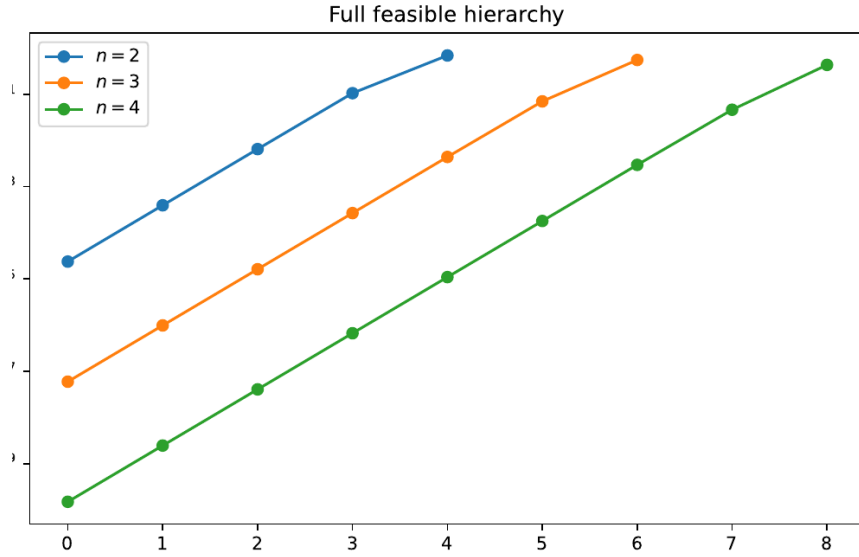


Figure 1: Worst centered relative error for the full feasible hierarchies $0 \leq k \leq 2n$, $n = 2, 3, 4$. The final point of each curve is the corresponding Gauss–Legendre rule.

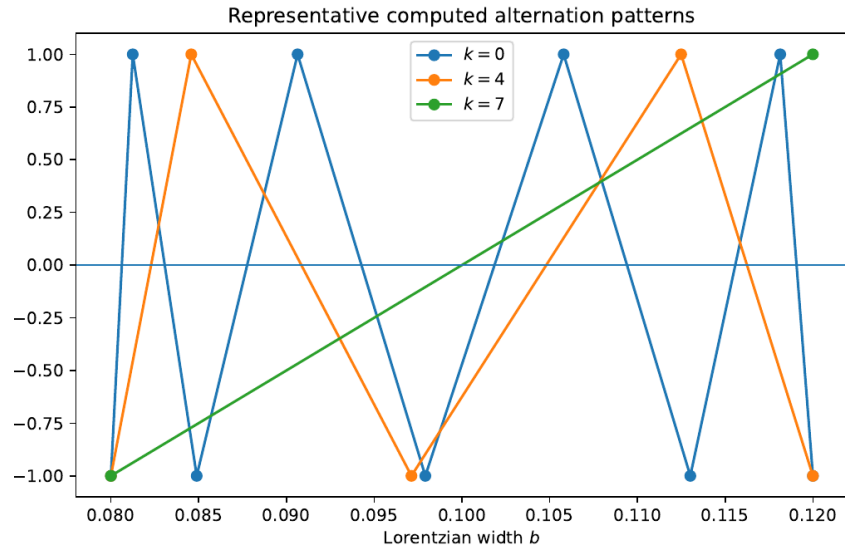


Figure 2: Representative normalized computed alternation patterns from the lower endpoint ($k = 0$), midpoint ($k = 4$), and upper half ($k = 7$) of the $n = 4$ hierarchy.

6. Computer-assisted existence and upper-half minimax certification

For each upper-half pair

$$(2, 3), (3, 4), (3, 5), (4, 5), (4, 6), (4, 7),$$

put $d = 2n - k$, fix $b_0 = 0.08$ and $b_d = 0.12$, and use the unknown vector

$$z = (y_1, \dots, y_n, w_1, \dots, w_n, b_1, \dots, b_{d-1}, E).$$

It has $4n - k$ components. The square system consists of the k moment equations, $d + 1$ signed equal-error equations, and $d - 1$ stationarity equations. The signs are fixed by the ordered alternation data and begin with negative error at $b = 0.08$. The earlier alternation theorem is written in $s = b^2$; because $b \mapsto b^2$ is strictly increasing and $2b > 0$ on the structural interval, the ordered extrema, uniform norm, and stationarity conditions are equivalent in the two coordinates.

Starting from the published decimals, Newton iteration at 180 decimal digits produces a center z_0 . Let X be the coordinatewise ball of absolute radius 10^{-12} about that center, let Y be a decimal point preconditioner computed from $F'(z_0)^{-1}$, and define the Krawczyk image

$$K(z_0, X) = z_0 - YF(z_0) + \{I - YF'(X)\}(X - z_0).$$

All algebraic and transcendental interval evaluations use Arb with 180 decimal digits. The inclusion argument follows the standard interval Newton–Krawczyk framework [16, 17], with Arb providing outward-rounded midpoint-radius arithmetic [18]. Table 2 reports the two principal checks. The componentwise inclusion margin is the minimum distance from K_i to the corresponding boundary of X_i .

Table 2: Krawczyk certificates for the six upper-half square systems.

(n, k)	Unknowns	Int. ext.	Norm upper	Inclusion margin
(2, 3)	5	0	1.741e-09	9.99986e-13
(3, 4)	8	1	2.741e-08	9.99999e-13
(3, 5)	7	0	3.432e-09	9.99991e-13
(4, 5)	11	2	5.823e-07	9.99997e-13
(4, 6)	10	1	4.376e-08	9.99999e-13
(4, 7)	9	0	3.439e-08	9.99995e-13

In every case, $K(z_0, X) \subset \text{int } X$ and the displayed matrix norm is below one. Hence the fixed-sign square system has one and only one exact zero in the stated box. Direct interval margins also verify

$$0 < y_1 < \cdots < y_n < 1, \quad w_j > 0,$$

ordered internal extrema in $(0.08, 0.12)$, and $E > 0$. This is local uniqueness of a system root inside the certified box; it is not a claim of global uniqueness among all minimizers.

A second interval audit closes the remaining supremum condition in the alternation theorem. For each internal extremum, the second derivative of the relative error has the sign opposite to the extremal error on a radius- 10^{-5} neighborhood. On the complementary subintervals, adaptive Arb subdivision proves the expected strict sign of the first derivative. The largest audit, $(n, k) = (4, 5)$, visits 6697 derivative subintervals; its smallest signed derivative margin is 7.55×10^{-8} , while its smallest curvature margin is 3.39. Together with the exact equations at the certified zero, these signs show that the listed alternating value E is the global uniform relative-error norm on $[0.08, 0.12]$.

Proposition 8 (Computer-assisted upper-half certification). *For each of the six pairs above, there exists a unique zero of the specified fixed-sign square system in the reported radius- 10^{-12} box. Its nodes are ordered, its weights are positive, its internal extrema are ordered, and its signed level is nonzero. Moreover, the relative error has global norm E and the required $2n - k + 1$ -point alternation. Consequently, the associated exact rule admits no strictly better competitor in $\mathcal{A}_{n,k}$.*

Proof. The Krawczyk inclusion and the norm bound prove existence and uniqueness of the system zero in each box and nonsingularity of every interval Jacobian covered by the certificate. The explicit coordinate margins prove ordering and positivity. The interval first- and second-derivative signs establish monotonicity between consecutive certified extrema and the correct local maximum/minimum orientation inside each extremum neighborhood. Since the square-system equations give the alternating values $\pm E$ exactly, no larger absolute error occurs on the structural interval. The final statement follows from the class-specific alternation theorem. $\square$

The machine-readable boxes, residual intervals, derivative-segment audits, source code, input hashes, and exact rerun commands accompany the

manuscript. The certificate is for the reconstructed mathematical systems and does not imply global uniqueness outside the certified boxes.

A reproducibility audit subsequently identified a non-rigorous center-generation detail in the archived script: the two `mpmath` endpoint constants had been constructed at import-time default precision before the working precision was raised. This affected only the diagnostic Newton center, not the rigorous system, because every Arb evaluation used the exact decimal endpoints 0.08 and 0.12. Reconstructing the endpoints after setting precision reduced the maximum point residual to below 2×10^{-180} , and the corrected script again certified all six boxes.

7. Independent reproduction and cross-artifact audit

A separate verification path was written without importing the Round 50 certifier and without copying its handwritten analytic Jacobian. It forms the square-system Jacobian by forward automatic differentiation, first over high-precision point arithmetic and then over Arb intervals. Applied to the precision-corrected centers and the same frozen input data, it independently reproduces strict Krawczyk inclusion for all six upper-half boxes. The largest reproduced bound on $\|I - YF'(X)\|_\infty$ is below 6.7×10^{-7} . The term “independent” here refers to an author-produced independent code path, not to third-party replication.

The same audit evaluates every frozen rule in the complete hierarchy

$$(n, k), \quad n = 2, 3, 4, \quad 0 \leq k \leq 2n.$$

All 21 operators pass ordering, positivity, moment, supplied alternation-point, and stationarity checks. Across the hierarchy, the largest reproduced moment residual is 6.28×10^{-16} , the largest supplied-extremum value discrepancy is 5.54×10^{-15} , and the largest internal-stationarity residual is 1.48×10^{-13} . A 50,001-point dense screen provides an additional numerical continuum check but is not used as a proof of minimaxity or root existence.

The Round 49 interval selector was rerun from its portable certifier and reproduced all reported counts and the certified fraction 0.9959563612937927. Repository-wide consistency tests verify the hashes of the Round 43 inputs used by Round 50, agreement of the three upper-half certificate paths, and agreement between generated data and reported manuscript constants. In the frozen audit environment, 35 of 35 cross-artifact checks and 7 of 7 automated

regression tests pass. These checks strengthen reproducibility but do not replace external specialist review or an execution on a separately provisioned machine.

8. Exact coefficient-space rule selection

For $C \geq 0$ and $\rho > 1$, define

$$\mathcal{H}(C, \rho) = \left\{ h : h(x) = \sum_{j=0}^{\infty} a_j T_j(x), \quad |a_j| \leq C \rho^{-j} \right\}.$$

Because $|T_j(x)| \leq 1$ on $[-1, 1]$ and $\sum_j \rho^{-j} < \infty$, every series in this definition converges absolutely and uniformly and therefore represents a continuous function.

For a fixed bounded quadrature functional Q , define

$$\Delta_j(Q) = I[T_j] - Q[T_j]$$

and

$$G_Q(\rho) = \sum_{j=0}^{\infty} \rho^{-j} |\Delta_j(Q)|.$$

The series converges because

$$|\Delta_j(Q)| \leq \|I - Q\|_{C[-1,1] \rightarrow \mathbb{R}}.$$

Theorem 9 (Exact coefficient-box norm). *For every fixed bounded quadrature functional Q ,*

$$\sup_{h \in \mathcal{H}(C, \rho)} |I[h] - Q[h]| = CG_Q(\rho).$$

Proof. Uniform convergence permits termwise application of the bounded functional $I - Q$, so

$$(I - Q)h = \sum_{j=0}^{\infty} a_j \Delta_j(Q).$$

The coefficient bounds give

$$|(I - Q)h| \leq C \sum_{j=0}^{\infty} \rho^{-j} |\Delta_j(Q)|.$$

Equality is attained by choosing

$$a_j = C\rho^{-j} \operatorname{sign} \Delta_j(Q),$$

with any value in $[-C\rho^{-j}, C\rho^{-j}]$ when $\Delta_j(Q) = 0$. The resulting series belongs to $\mathcal{H}(C, \rho)$. $\square$

For $c_{\max} \geq 0$, define the parameterized mixed class

$$\mathcal{F}(C, \rho, c_{\max}) = \{h + cL_b : h \in \mathcal{H}(C, \rho), |c| \leq c_{\max}, b \in B\}.$$

For a fixed rule, let

$$S_Q = \max_{b \in B} |I[L_b] - Q[L_b]|.$$

The maximum exists because B is compact and the error is continuous in b .

Theorem 10 (Exact mixed-class risk). *For every fixed bounded quadrature functional Q ,*

$$\sup_{f \in \mathcal{F}(C, \rho, c_{\max})} |I[f] - Q[f]| = CG_Q(\rho) + c_{\max} S_Q.$$

Proof. The upper bound follows from the triangle inequality and the preceding theorem. Choose $b_{\star} \in B$ attaining S_Q . Select $h_{\star} \in \mathcal{H}(C, \rho)$ such that

$$(I - Q)h_{\star} = CG_Q(\rho),$$

and choose

$$c_{\star} = c_{\max} \operatorname{sign} (I[L_{b_{\star}}] - Q[L_{b_{\star}}]).$$

Then both error contributions have the same sign and their absolute values add, proving attainability. $\square$

Let $Q_0, \dots, Q_{2n}$ be the finite hierarchy of candidate rules. When $C > 0$, put $\tau = c_{\max}/C$. The exact worst-case selector within this candidate set is

$$k^{\star}(\rho, \tau) \in \arg \min_{0 \leq k \leq 2n} \{G_{Q_k}(\rho) + \tau S_{Q_k}\}.$$

Remark 11. This selector is exact for the stated coefficient box and structural family, but only among the precomputed candidates $\{Q_0, \dots, Q_{2n}\}$. It does not assert global optimality over all quadrature rules. The rules were constructed by a relative structural minimax objective, whereas S_Q is the absolute structural error appropriate to the additive mixed class.

Table 3: Full $n = 4$ structural and smooth-risk trade-off at $\rho = 1.5$.

k	Structural screening margin S_k	$G_k(1.5)$
0	5.55627e-09	9.11242e-01
1	9.17189e-08	3.95465e-01
2	1.50911e-06	2.04657e-01
3	2.48007e-05	9.59856e-02
4	4.07341e-04	4.49242e-02
5	6.68824e-03	2.14018e-02
6	1.09706e-01	1.17665e-02
7	1.71386e+00	6.62192e-03
8	1.60298e+01	3.42291e-03

9. Computer-assisted interval certification for the rounded rules

The exact risk theorems apply to each fixed operator. For the published decimal operators $\tilde{Q}_k$, we treat every displayed decimal node and weight as an exact decimal input enclosed by ball arithmetic. All transcendental and algebraic evaluations in the structural stage use 80-decimal-digit Python-FLint/Arb balls [18].

For each $k = 0, \dots, 8$, let

$$e_k(b) = I[L_b] - \tilde{Q}_k[L_b].$$

The interval $[0.08, 0.12]$ is adaptively bisected. On each subinterval X , two enclosures are computed: the natural interval extension $e_k(X)$, and the mean-value enclosure

$$e_k(m) + e'_k(X)(X - m),$$

where m is the midpoint. The smaller valid absolute upper bound is used in a branch-and-bound search. Point evaluations provide lower witnesses. This produces rigorous enclosures

$$S_k = \max_{b \in [0.08, 0.12]} |e_k(b)| \in [\underline{S}_k, \overline{S}_k].$$

All nine largest lower witnesses occur at $b = 0.08$; the maximum relative enclosure width is

$$1.437e - 10.$$

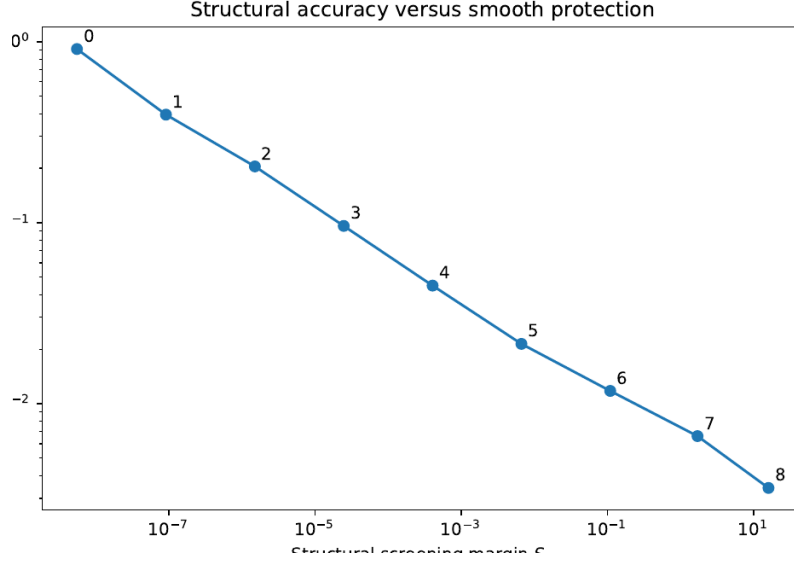


Figure 3: Each added moment decreases the smooth-risk norm but increases the Lorentzian structural error. Labels indicate the moment depth k .

For the smooth component, each Chebyshev defect

$$\Delta_j(\tilde{Q}_k) = I[T_j] - \tilde{Q}_k[T_j]$$

is enclosed by Arb through degree $J = 500$. Since

$$|\Delta_j(\tilde{Q}_k)| \leq \|I - \tilde{Q}_k\| \leq 2 + 2 \sum_i w_i,$$

the remaining tail is bounded by

$$\sum_{j>J} \rho^{-j} |\Delta_j| \leq \left(2 + 2 \sum_i w_i\right) \frac{\rho^{-(J+1)}}{1 - \rho^{-1}}.$$

The nonnegative finite sums are evaluated by outward-rounded Horner recurrences. On a cell

$$[\rho_-, \rho_+] \times [t_-, t_+], \quad t = \log_{10} \tau,$$

monotonicity gives rigorous lower and upper risk bounds from ρ_+, ρ_- and $10^{t-}, 10^{t+}$, respectively. A cell is assigned to k only when the upper bound for k is strictly below the lower bound for every competitor.

Adaptive binary subdivision to depth $d = 24$ visits 344,615 cells and certifies a unique winner on

$$99.595636\%$$

of the rectangular coordinate area

$$(\rho, \log_{10} \tau) \in [1.05, 5] \times [-12, 12].$$

The remaining

$$0.404364\%$$

is unassigned and lies near transition curves. The percentage is an area fraction in these coordinates, not a probability statement.

Table 4: Fraction of the interval-certified coordinate area assigned to each moment depth.

k	Fraction of certified area
0	21.993%
1	9.179%
2	8.320%
3	8.493%
4	8.536%
5	8.572%
6	8.312%
7	7.054%
8	19.540%

The certificate applies to the rounded operators, coefficient box, structural interval, and coordinate domain stated above. It is logically separate from the upper-half root certificates in Section 6: the selector uses the published decimal operators, whereas the Krawczyk calculations establish exact roots only for the six upper-half nonlinear systems.

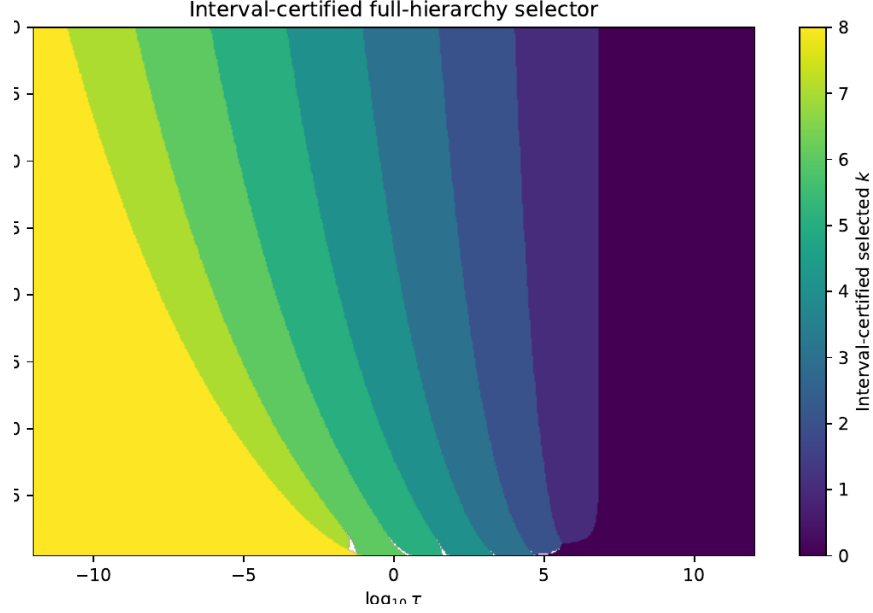


Figure 4: Interval-certified full-hierarchy selector. Blank regions remain unresolved because candidate risk intervals overlap.

10. External validity and reconstruction

10.1. Small pole-location mismatch

Without retraining, the centered rules were tested on

$$L_{a,b}(x) = \frac{1}{(x-a)^2 + b^2}, \quad |a| \leq 0.03, \quad 0.08 \leq b \leq 0.12.$$

Table 5: Worst relative errors on the shifted-Lorentzian stress-test box.

Method	Evaluations	Worst error	Worst $ a $	Worst b
PMC-k0	8	4.522336e-06	0.0300	0.0800
PMC-k4	8	1.025564e-03	0.0300	0.0845
Composite-Simpson-9	9	7.426762e-02	0.0300	0.1200
Gauss-Legendre-8	8	4.300499e-01	0.0000	0.0800

The $k = 0$ rule has worst relative error 4.52×10^{-6} , whereas the Gauss endpoint has worst error 4.30×10^{-1} on the same box. The centered error is

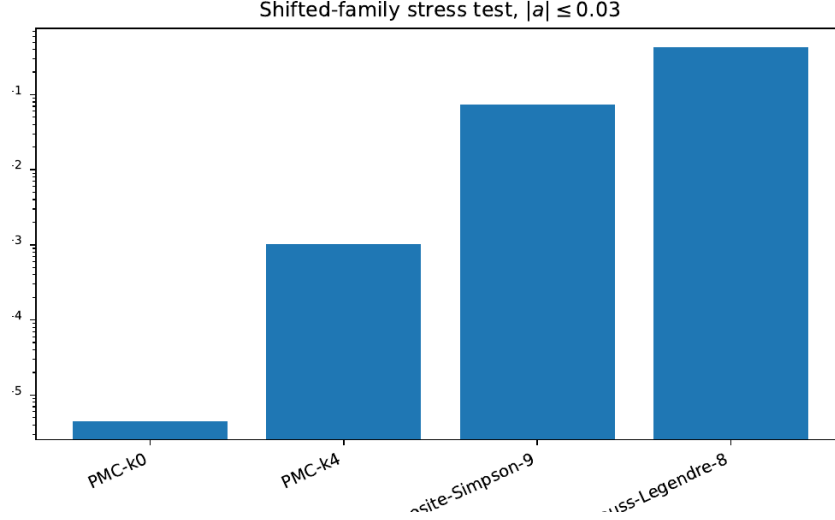


Figure 5: Stress test under small pole-location mismatch. The comparison supports local structural robustness, not shifted-family minimaxity or universal superiority.

nevertheless inflated by more than four orders of magnitude, so the correct interpretation is local robustness rather than shift invariance.

10.2. Multi-start reconstruction

Using the same equioscillation equations and solver implementation, the six upper-half problems

$$(2, 3), (3, 4), (3, 5), (4, 5), (4, 6), (4, 7)$$

were reconstructed from twelve perturbed starts each. A run was accepted only when nodes were ordered, weights were positive, moment residuals were below 10^{-10} , and scaled equioscillation residuals were below 10^{-8} .

All accepted solutions in each problem formed a single error-level cluster. No accepted start found a better rule by more than 10^{-6} relative, and the maximum best-rule error difference was 2.38×10^{-12} . This is a sensitivity test with perturbed initializations. It reduces the risk of an isolated local numerical artifact, but it is neither an independent implementation nor a proof of global uniqueness.

Table 6: Multi-start reconstruction of the upper-half rules. Error differences are relative to the published minimax level.

(n, k)	Accepted starts	Error difference	Max. node difference	Max. weight difference
(2, 3)	12/12	2.64e-16	5.55e-17	5.55e-17
(3, 4)	10/12	6.78e-14	4.44e-16	5.00e-16
(3, 5)	12/12	5.31e-14	1.29e-14	1.32e-14
(4, 5)	8/12	2.38e-12	4.44e-15	4.27e-15
(4, 6)	11/12	9.22e-14	1.03e-14	1.05e-14
(4, 7)	12/12	5.11e-13	8.95e-14	6.43e-14

11. Relation to prior work and novelty scope

11.1. Structure-aware and generalized quadrature

Gautschi’s rational Gauss-type rules and later survey establish the principle that a quadrature formula can integrate rational functions selected to match important external poles [1, 2]. The present work therefore does not introduce pole-aware quadrature. Its structural objective is different: it minimizes relative error over a continuum of Lorentzian widths instead of enforcing exactness at a finite set of prescribed poles. Related Gauss-type formulas with external denominator zeros further illustrate the breadth of rationally structured quadrature [11].

Generalized Gaussian quadrature for Chebyshev systems supplies positive interior-node rules exact for non-polynomial spaces, and complete Chebyshev systems support stepwise increases in exactness [3]. Positive exact formulas, prescribed-node moment constructions, and positive-node geometry are also well developed [7, 8, 15]. Accordingly, neither positivity nor an ordered hierarchy of exactness is claimed as new. Here the moment depth indexes a family selected by a continuum Stieltjes minimax criterion at fixed node-pair count.

Optimization-based nested quadrature shows that nonlinear moment-matching systems can be solved flexibly [9]. Our nonlinear solver and multi-start study are reproducibility mechanisms, not the principal conceptual contribution.

11.2. Rational approximation and minimax constraints

Rational minimax approximation and equioscillation are classical; adaptive barycentric algorithms make high-order problems computationally accessible [4]. Interpolation-constrained rational minimax approximation already includes prescribed function values and higher-order derivative conditions [5].

After the change of variable $z = 1/s$, matching Laurent coefficients at infinity is closely related to confluent interpolation at $z = 0$. Thus the novelty claim cannot be “minimax plus constraints.” The alternation theorem in this paper is instead a concise specialization to positive Stieltjes-response quadrature, where the shared coefficients reduce the difference-numerator degree.

Markov-function rational approximation provides relevant relative-error context [10]. Horning and Trefethen make the link from approximation to quadrature explicit by deriving quadrature from rational approximations of Cauchy transforms; their framework also connects Gauss quadrature with Padé approximation at infinity [6]. This is the closest conceptual comparator. The distinction here is the simultaneous enforcement of positive residues, a variable polynomial-moment prefix, a continuum relative-minimax objective, and an exact mixed-class selector.

11.3. Worst-case rule selection

Worst-case analysis for analytic function classes is an established part of quadrature theory [12, 13], and recent positive-weight kernel quadrature gives a different convex-geometric worst-case framework [14]. The coefficient-box identity used here is elementary duality. Its role is to make the smooth residual commensurate with the Stieltjes structural error, producing

$$G_k(\rho) + \tau S_k.$$

The paper’s practically distinctive output is therefore not a new norm identity in isolation, but a selector and interval-certified decision map over a finite family of positive structural rules.

11.4. Defensible novelty statement

The targeted audit found no direct source combining all of the following: positive symmetric Stieltjes residues, continuum Lorentzian relative minimax, variable moment depth at fixed node-pair count, an exact mixed smooth/structural selector, and a validated parameter partition. The defensible claim is this combined model-aware framework. The paper does not claim to invent positivity, moment matching, structure-aware quadrature, constrained rational minimax, equioscillation, or the Gauss/Padé connection separately.

12. Limitations

The theoretical and numerical study is one-dimensional and symmetric. The principal structural family is the centered Lorentzian kernel. The shifted-family experiment is a stress test and is not itself minimax-optimized.

The six upper-half fixed-sign systems have computer-assisted existence and local uniqueness boxes, together with interval proofs that their alternating levels are global on the stated structural interval. This does not prove global uniqueness among equal-norm minimizers, and it does not provide interval root certificates for the lower-half rules. The multi-start study uses the original nonlinear equations and solver. The later forward-AD/Arb verifier is an independent author-produced code path, but neither calculation is a third-party replication. Moreover, the global derivative/curvature audit has been rerun rather than reimplemented independently.

The interval-certified selector covers 99.5956% of the chosen $(\rho, \log_{10} \tau)$ -coordinate rectangle and leaves 0.4044% unassigned. The coverage depends on the specified coefficient box, Lorentzian interval, candidate rules, coordinate domain, truncation degree $J = 500$, and subdivision depth. It is not a probability statement and does not imply optimality over quadrature rules outside the finite hierarchy.

The regression suite has been exercised in the frozen development environment, but a clean offline installation on a separately provisioned machine remains a release-stage check. The empirical near-geometric price per moment is not an asymptotic theorem. The literature audit is targeted rather than exhaustive, and equivalent formulations may exist under moment-space, generalized Gaussian, constrained Markov, or interpolatory rational terminology. The manuscript therefore avoids “first” and “unprecedented” claims.

13. Conclusion

Within the stated positive symmetric Stieltjes class, the variable moment depth organizes a path

$$k = 0 \quad \longrightarrow \quad k = 2n$$

from a structural minimax rule to Gauss–Legendre. The feasible-range characterization and difference-degree argument support the numerical construction, while the exact mixed risk turns the finite hierarchy into a model-aware selector. Complete computations for $n = 2, 3, 4$, computer-assisted existence

and global-alternation certificates for the six upper-half rules, interval certification of the rounded-operator decision map, a shifted-pole stress test, and cross-artifact reproduction quantify both the promise and the limits of the approach.

The main contribution is not any ingredient separately. It is the combination of positive Stieltjes realization, continuum structural minimax, variable polynomial protection, exact mixed-model selection, and computer-assisted certification in one reproducible framework. The principal remaining scientific gate is external specialist review against the closest generalized-quadrature, moment-space, and constrained-rational literature. Extensions beyond symmetry and the centered kernel are future work rather than requirements for the stated finite-class contribution.

Data and code availability

A versioned research archive accompanying this manuscript contains the published rules, generated data, interval certificates, source code, regression tests, dependency lock file, and SHA-256 manifest needed to reproduce the reported results. Before formal submission, this frozen archive will be deposited in a public repository that issues a persistent identifier, and the identifier will be inserted here. The existing public GitHub repository currently records an earlier project baseline and is not yet the archival record for the present manuscript version.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with literature organization, code prototyping, manuscript structuring, and language revision. The author reviewed and verified the mathematical derivations, numerical results, references, interpretations, and final wording and takes full responsibility for the content of the article.

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